

PHATTHALUNG AGROMET, Thailand

WMO: 485600

Lat: 7.581N Lon: 100.166E Elev: 4 StdP: 101.28 Time Zone: 7.00 (E07) Period: 08-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 22.7	(c) 23.3	(d) 20.7	(e) 15.3	(f) 28.5	(g) 21.3	(h) 15.9	(i) 27.9	(j) 6.0	(k) 29.5	(l) 5.2	(m) 28.7	(n) 0.7	(o) 240	(p) 0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 7.7	(c) 35.1	(d) 27.0	(e) 34.3	(f) 26.8	(g) 33.7	(h) 26.6	(i) 28.4	(j) 33.4	(k) 28.0	(l) 32.8	(m) 27.6	(n) 32.2	(o) 2.6	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.7	31.7	26.6	22.2	31.1	26.3	21.7	30.6	91.5	33.8	89.4	33.0	87.9	32.6	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.4	4.4	3.7	21.3	35.9	1.2	1.0	20.5	36.6	19.8	37.2	19.1	37.8	18.2	38.6		
			20.6	29.3	1.1	1.2	19.8	30.2	19.1	30.9	18.5	31.6	17.6	32.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	26.9	27.5	28.3	29.2	29.3	28.8	28.6	28.6	28.3	27.9	27.1	26.8
	DBStd	1.33	1.06	1.00	0.97	1.16	1.06	1.02	0.94	0.94	0.98	1.07	1.21	1.08
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6614	524	491	568	576	598	563	576	577	551	556	514	519
	CDD18.3	3572	266	258	310	326	339	313	317	318	301	298	264	261
	CDH23.3	37898	2540	2726	3517	3876	3903	3525	3462	3432	3108	2950	2492	2368
	CDH26.7	14117	667	923	1422	1719	1677	1467	1412	1397	1168	1008	704	552
Wind	WSAvg	1.1	1.1	1.2	1.1	1.1	1.0	1.2	1.3	1.3	1.3	1.0	0.8	0.9
Precipitation	PrecAvg	2092	84	43	94	115	148	137	145	174	176	268	389	320
	PrecMax	2715	324	250	517	275	307	220	232	357	249	449	745	817
	PrecMin	1515	5	1	0	20	78	31	67	72	108	164	190	78
	PrecStd	324	81	53	100	59	43	47	40	60	40	65	150	169
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	32.5	34.2	36.2	36.2	35.0	34.7	35.0	34.5	34.1	32.5	31.7
		MCWB	27.2	26.7	27.3	28.7	28.1	26.1	25.4	25.7	25.7	26.1	26.9	26.9
	2%	DB	31.0	31.7	33.3	35.0	35.0	34.1	33.8	34.0	33.5	33.0	31.6	30.9
		MCWB	26.5	26.3	26.9	27.9	27.7	26.4	26.0	25.8	26.0	26.3	26.8	26.6
	5%	DB	30.3	31.3	32.6	33.9	33.9	33.3	33.1	33.2	32.7	32.0	30.9	30.2
		MCWB	26.0	26.1	26.7	27.6	27.3	26.5	26.0	25.9	26.1	26.5	26.7	26.3
	10%	DB	29.7	30.7	31.9	33.0	33.0	32.6	32.2	32.3	31.8	31.2	30.2	29.5
		MCWB	25.7	25.9	26.4	27.3	27.1	26.4	26.0	25.9	26.0	26.3	26.5	26.0
	0.4%	WB	27.7	27.6	28.0	29.0	29.0	28.2	27.7	27.9	27.7	27.7	27.8	27.7
		MCDB	31.1	31.6	33.0	34.9	34.2	33.0	32.5	32.7	32.4	31.6	31.1	30.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.0	27.0	27.5	28.4	28.3	27.6	27.1	27.2	27.2	27.2	27.3	27.1
		MCDB	30.4	31.0	32.3	33.9	33.6	32.4	31.9	31.9	31.7	31.2	30.6	30.1
	5%	WB	26.5	26.5	27.1	27.9	27.8	27.2	26.7	26.7	26.7	26.8	27.0	26.6
		MCDB	29.6	30.4	31.8	33.1	32.9	31.8	31.3	31.3	31.0	30.7	30.4	29.5
	10%	WB	26.0	26.1	26.7	27.4	27.4	26.8	26.3	26.3	26.2	26.4	26.6	26.2
		MCDB	29.0	29.9	31.0	32.2	32.0	31.1	30.8	30.7	30.3	30.2	29.7	28.8
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.9	7.4	7.7	7.8	7.7	7.9	8.0	7.4	6.7	5.2	4.8
		MCDBR	6.4	7.4	8.5	9.2	9.1	8.9	9.0	9.1	8.6	8.1	6.7	6.3
		MCWBR	2.9	3.1	3.3	3.5	3.3	2.9	2.9	2.8	2.8	2.9	3.0	2.8
	5% WB	MCDBR	6.1	6.7	7.7	8.6	8.5	8.1	8.3	8.4	7.9	7.2	6.3	5.8
		MCWBR	3.0	3.2	3.4	3.6	3.5	3.3	3.3	3.4	3.3	3.1	3.1	2.9
Clear-Sky Solar Irradiance	taub		0.473	0.475	0.482	0.464	0.447	0.450	0.454	0.450	0.441	0.445	0.460	0.470
	taud		2.249	2.225	2.234	2.292	2.369	2.390	2.367	2.363	2.383	2.379	2.329	2.277
	Ebn at Noon		844	854	848	851	848	834	834	851	871	870	850	837
	Edh at Noon		140	147	147	137	123	119	123	126	125	124	128	134
All-Sky Solar Radiation	RadAvg		4.77	5.74	5.88	5.78	5.18	4.96	4.92	5.06	4.94	4.57	4.04	3.99
	RadStd		0.39	0.32	0.46	0.30	0.28	0.25	0.24	0.18	0.18	0.33	0.37	0.47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	+0.21	+0.16	N/A	+0.37	+0.21	+0.23	N/A	N/A	+88	+89

Nomenclature: See separate page